Scalability & Future Plan

Date	11 July 2026
Team ID	SWETID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Marks

Scalability & Future Plan:

Current System Limitations:

S No	Limitations	Impact	Priority to Address (High/Medium/Low)
1	Single-threaded Local Execution: Runs on a basic local Flask development server.	Multiple simultaneous user submissions will lock or crash the server.	High
2	Volatile Session Storage: System memory handles active inputs without a persistent database backend.	All historical evaluation records and submission data are lost when the server restarts.	High
3	Hardcoded Conditional Logic: Risk evaluations depend completely on manual script thresholds.	Changing regulatory compliance standards or credit risk updates requires rewriting code.	Medium

Scalability Plan:

S No	Scalability Aspect	Current State	Proposed Upgrade/Solution
1	User Load	Restricted to a single local client.	Migrate to an enterprise web server setup using Gunicorn or Nginx reverse proxies to scale concurrent connections.
2	Data Storage	In-memory tracking with zero logging.	Connect a production-ready database like PostgreSQL or MongoDB to record applicant telemetry securely.

3	Performance	In-memory script calculations execution.	Implement multi-worker routing processes to maintain fast response intervals under load.
4	Security	Unencrypted local HTTP form fields.	Enforce standard HTTPS (SSL/TLS) network layers and pass text inputs through server-side sanitization libraries.

Future Roadmap:

Phase	**Planned Feature/Enhancement**	**Target Timeline**	**Expected Impact**
Phase 1	Persistent Database Integration	1 Month	Retains client application histories for auditing and reporting purposes.
Phase 2	Machine Learning Core Upgrade	3 Months	Substitutes hardcoded script bounds with a dynamic classifier model to boost prediction accuracy.
Phase 3	Automated OCR Document Scanner	6 Months	Allows applicants to upload payslips or identity files directly to extract form data instantly.